



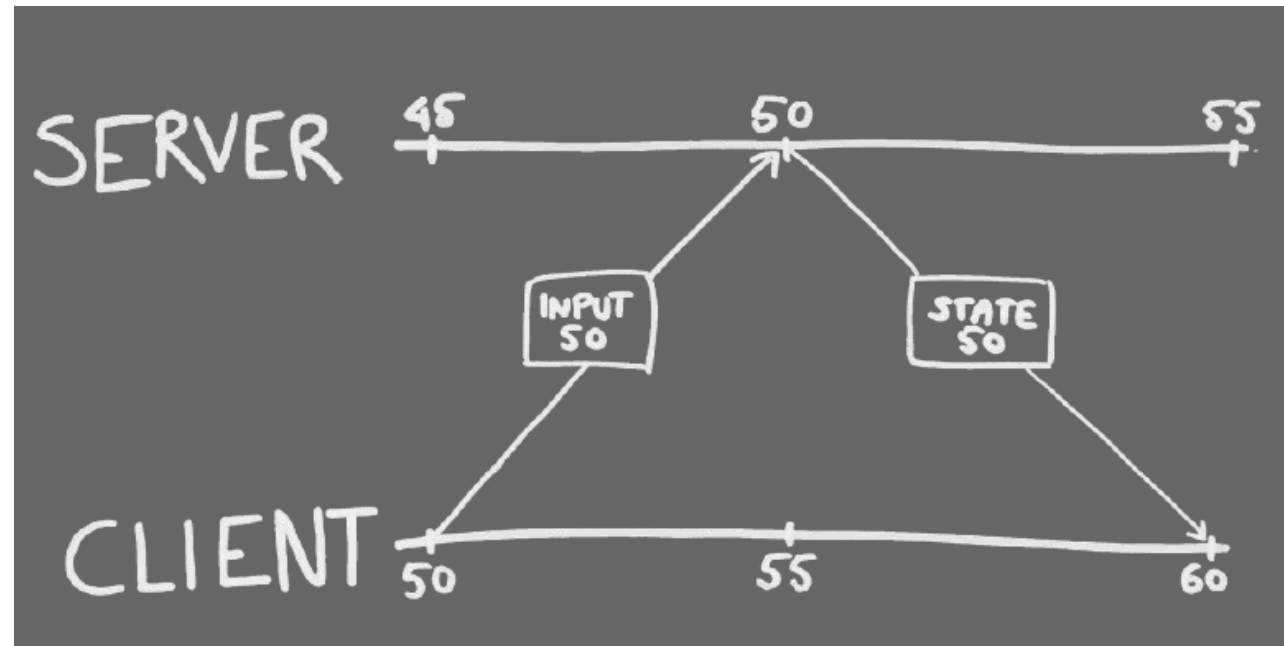
UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH
Centre de la Imatge i la Tecnologia Multimèdia

Lab Session 8: Reliability over UDP



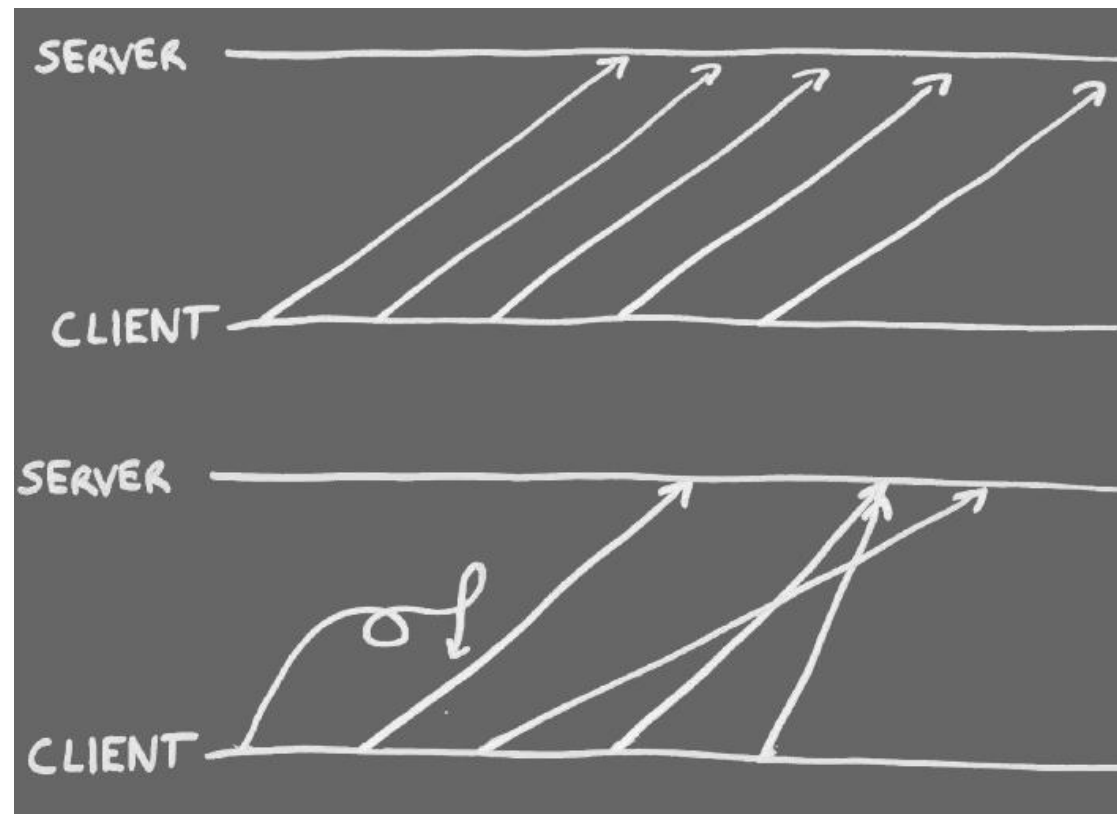
Real world conditions

- Lag is the delay provoked by the fact that data takes some time to travel and reach its destination.
- The time spend for a packet to be sent and receive some response is called RTT (round trip time)



Real World Conditions

- Ideally, packets would always reach its destination, and they would reach it in order (upper image).
- In the real world, however, some packets can be lost and even if they reach its destination, they can arrive out of order (lower image).



Real World Conditions

- Packet drop or packet loss: is when a packet never reaches its destination.
- Jitter: is the variance in time delay between data packets (e.g. because a routing reschedule) and it can provoke packets arriving unordered.

Methods to improve reliability

- Delivery notification: Used to ensure world state replication. Deals with packet loss and rearranging packets.
- Redundancy: Keep sending all inputs until server notifies reception

Delivery Notification System

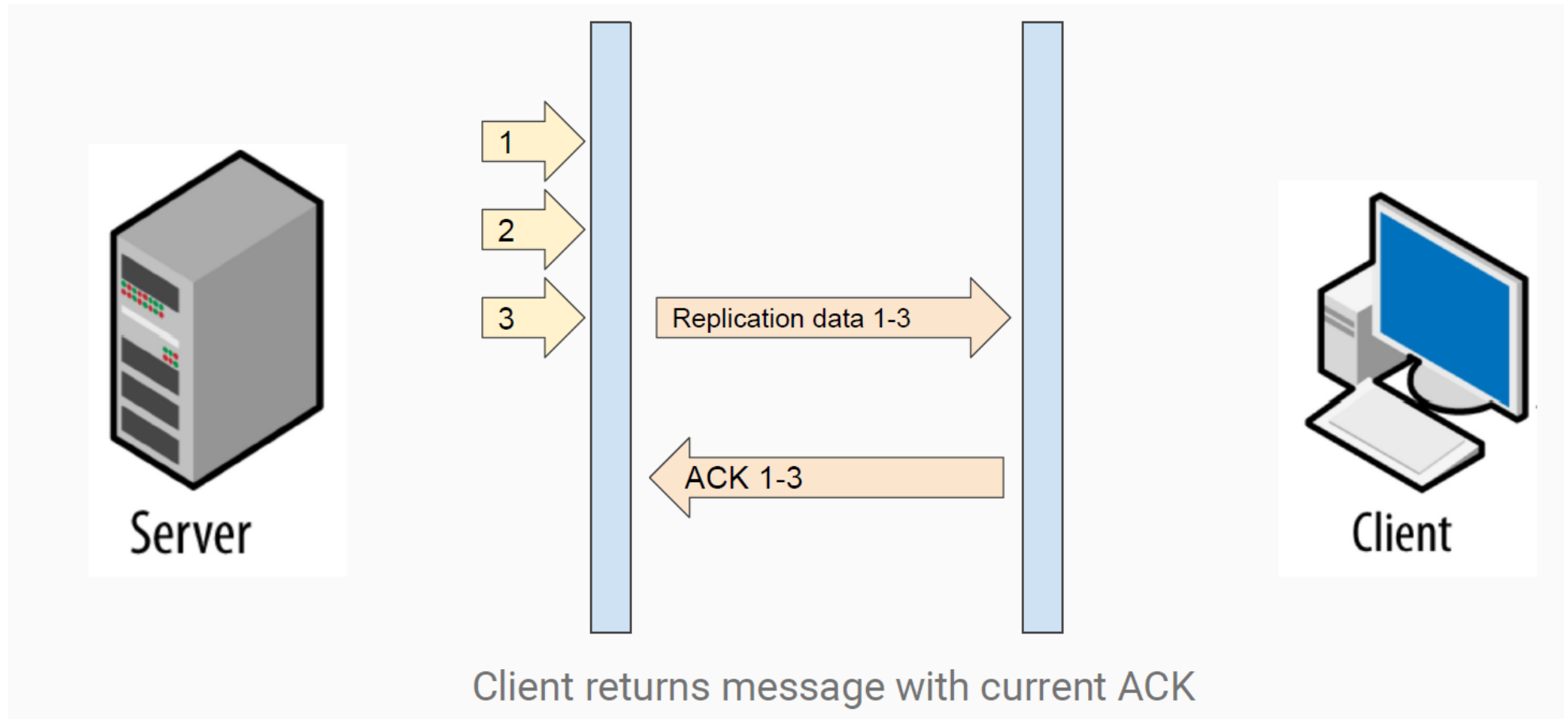
Core Concepts

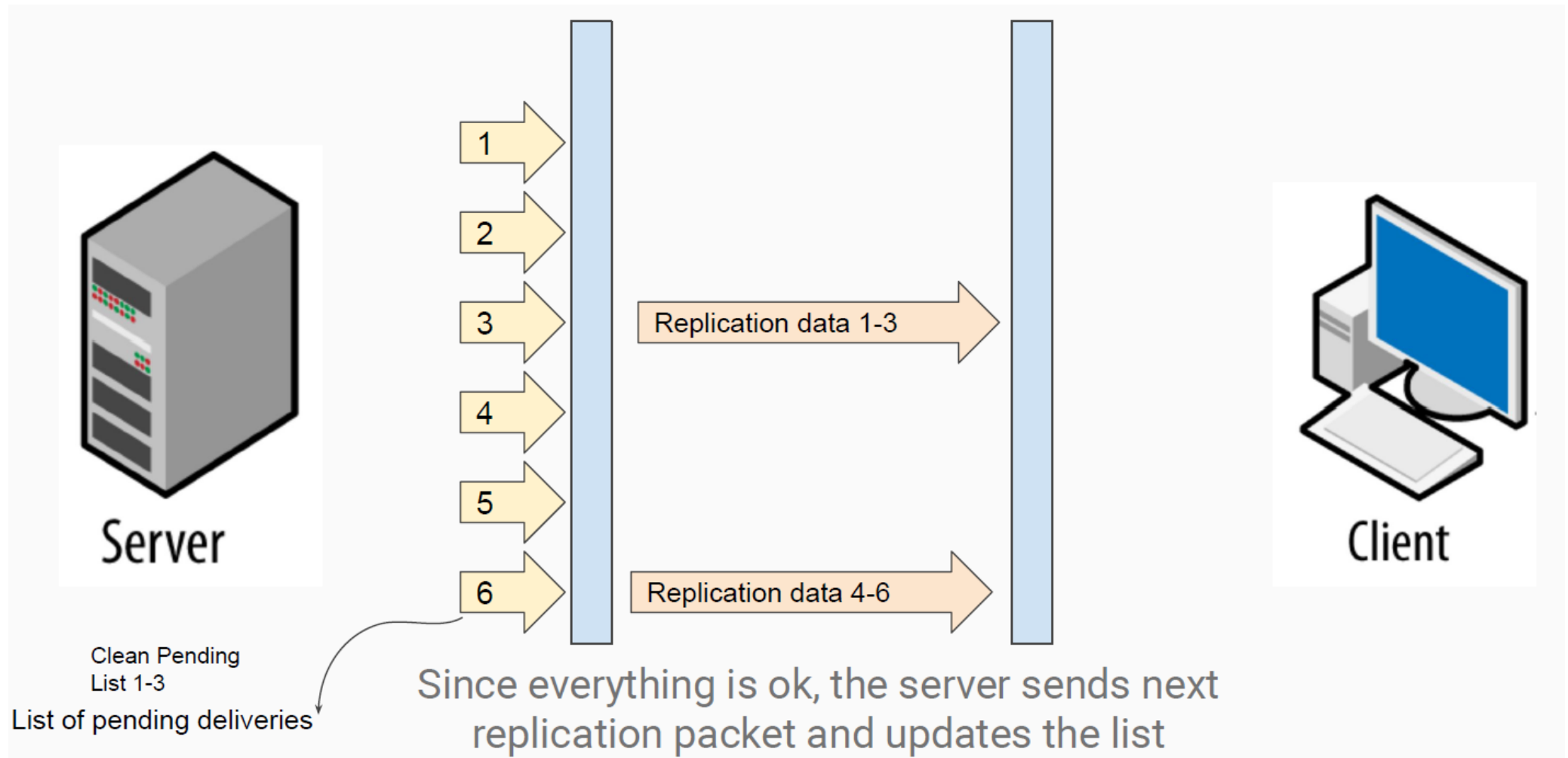
- Based around Acknowledgements (ACK)
- Signal between communicating processes to acknowledge the receipt of a message

Uses

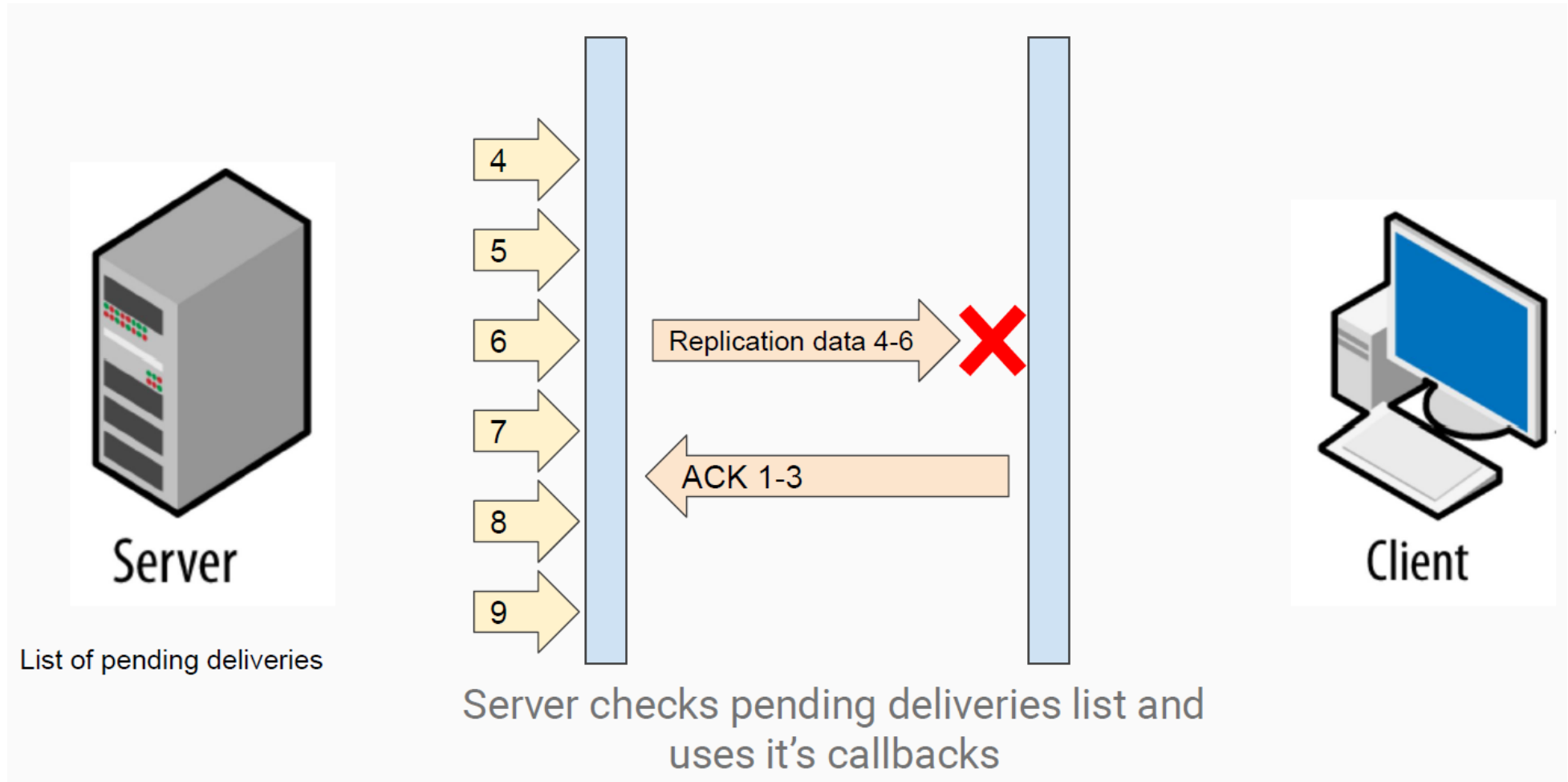
- Keeps track of the current state of communications
 - Lost packets
 - Tracks delivering order
 - Corrupted data
 - etc.

Usage case





Usage case



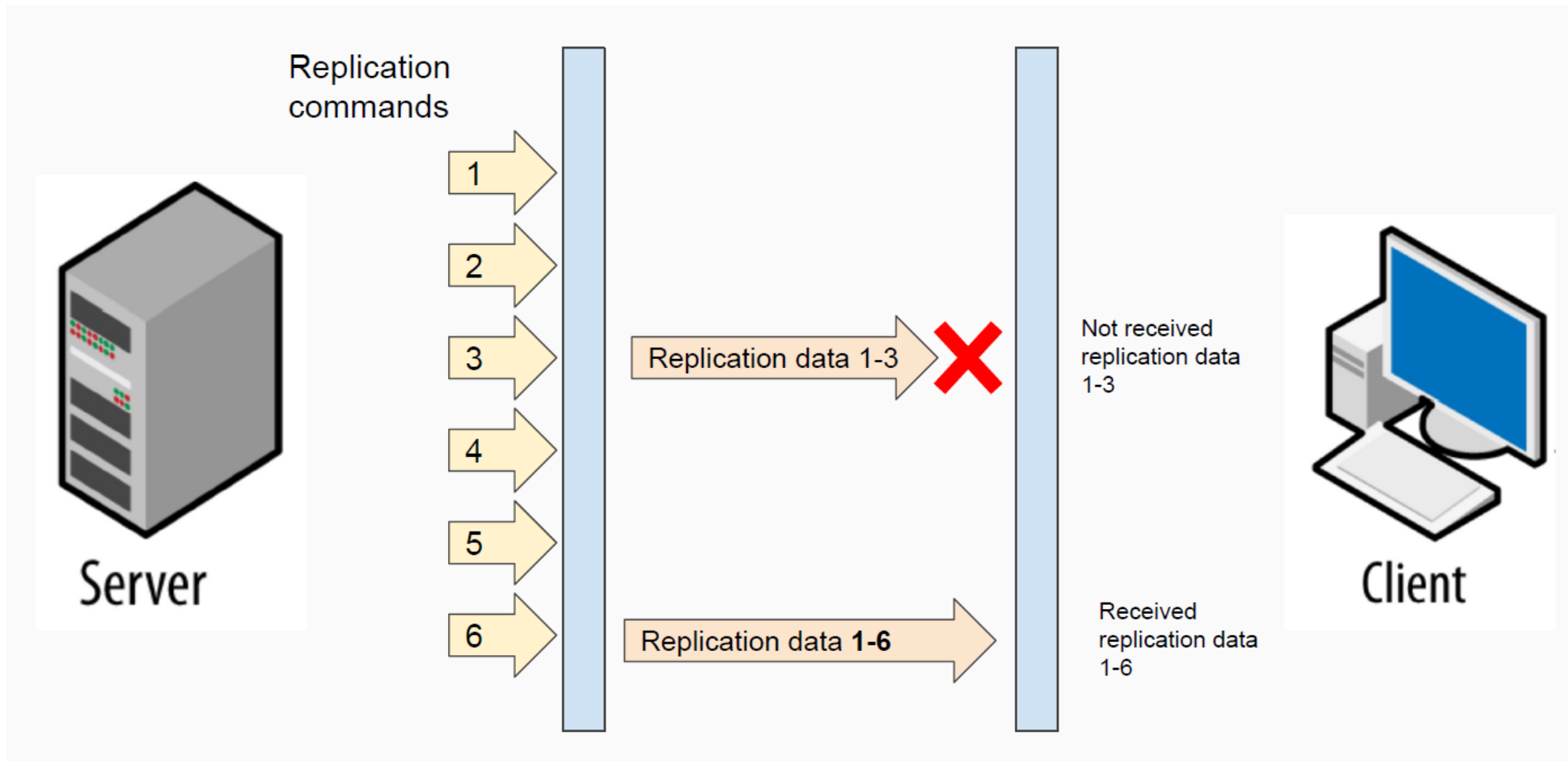
Callbacks

- Every delivery in our delivery list should have two.
 - On success: the client received this package in order and without any problems
 - Probably just remove the delivery from the list
 - On failure: the packet did not arrive correctly to the client.
 - Send it again (redundancy)
 - Discard it (priority)
 - Both → timeouts



Redundancy

Redundancy



Redundancy

- Used to recover lost packets
 - We store already sent packets in case we have to recover them
 - Keep in mind redundancy is mostly used for input packets
 - Smaller size. There's a limit to the amount of inputs a player can issue.

When do we send notifications?

- A couple of options
 - Each time we receive an input data packet
 - Faster response
 - If the client sends input data packets very frequently, we will use a lot of bandwidth
 - With the next replication packet
 - Slower response
 - Optimized bandwidth usage